When Does Private Equity Pay Off? US Private Equity vs Public Markets

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Technical terms are defined in the Glossary at the end of this note.

Executive summary

PE's edge sits with the managers, not with the asset class.

US private equity has beaten public markets over the long run, but on average it has not cleared the return hurdle investors set for it. Whether it pays off depends on access to the best managers, which most investors lack.

- **Long run:** over 25 years, the Cambridge Associates US PE Index returned 12.76% a year (to Mar 2026), against 8.74% for the S&P 500 and 6.18% for Baa corporate bonds (to Dec 2025).
- **Below the hurdle against the broad market:** measured like-for-like (mPME), PE beat the Russell 3000 by 2.7 points a year over 25 years, short of the c. 3-point margin many pension funds require, before any adjustment for buyout leverage.
- **A thin pass against small caps:** against the Russell 2000, closer in size to buyout targets, the 25-year premium is 3.7 points, because US small caps lagged large caps over the period. Neither index adjusts for leverage.
- **Vintage matters, but could not be timed:** value-add ranged from −3.0 to +15.1 points a year across the 1994–2013 vintages. It was highest when public markets did poorly during a fund's life; market returns before a fund's launch did not predict it.
- **Manager dispersion is a risk, not an edge:** on average, the top-quartile threshold beat the S&P 500 mPME by 11.5 points a year, while the bottom-quartile threshold trailed it by 2.7 points. Past performance is a weak guide to future results, so most investors cannot reliably pick the top quartile.

1. Question and approach

The question for an investor is not whether PE returns are high, but whether they beat what the same money would have earned in liquid markets. PE asks for a 10-year lock-up, high fees and limited transparency, so it must earn a premium over listed assets.

This note compares US private equity with US equities and bonds along three dimensions:

1. **Horizon:** annualised returns over 1 to 25 years.
2. **Vintage year:** performance by the year a fund started investing (1994–2013).
3. **Manager:** the spread between top- and bottom-quartile funds.

What this note adds. Its main findings echo established academic work: returns vary widely across funds and vintages (Kaplan & Schoar, 2005), and PE should be benchmarked with PMEs rather than raw IRRs (Harris, Jenkinson & Kaplan, 2014). This note is a data synthesis, not original research. It updates the evidence to March 2026, tests how the verdict depends on the benchmark (Russell 3000 vs Russell 2000), and checks whether good vintages could have been identified in advance.

2. Data and methodology

The analysis combines three public datasets. Every horizon return was rebuilt in an Excel model and cross-checked in Python with identical results; the model, notebooks and charts are on GitHub. Vintage-level data come from the September 2019 report, the latest free edition with vintage tables. Funds need about six years to mature, so current data would add only the 2014–2019 vintages.

Dataset	Content	As of
Cambridge Associates, US PE Benchmark Book Q1 2026	Pooled horizon IRR of 1,801 US private equity funds formed 1983–2026; mPME vs MSCI World/ACWI (constructed), Russell 3000 and Russell 2000	31 Mar 2026
Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019	Since-inception IRR, quartiles and mPME by vintage year, 1,366 funds (1994–2017)	30 Sep 2019
A. Damodaran, NYU Stern, Historical Returns	Annual total returns of the S&P 500, US small caps, T-bills, 10Y T-bonds and Baa bonds (1928–2025)	31 Dec 2025

Public assets are measured by their geometric annualised return (CAGR), which compounds the annual returns over the last *n* years:

CAGR = [(1 + r_1) × (1 + r_2) × … × (1 + r_n)]^(1/n) − 1, where r_t is the total return in year t.

PE is measured by its internal rate of return (IRR), net of fees, expenses and carried interest. The IRR is money-weighted: it depends on when capital is called and returned. Comparing it directly with a time-weighted CAGR is therefore *naive*.

The modified Public Market Equivalent (mPME) corrects this. Cambridge Associates invests each fund's capital calls in a public index at the same dates, and sells the same proportion of holdings at each distribution. The PE *value-add* is the fund's IRR minus the IRR of this shadow portfolio: a like-for-like comparison.

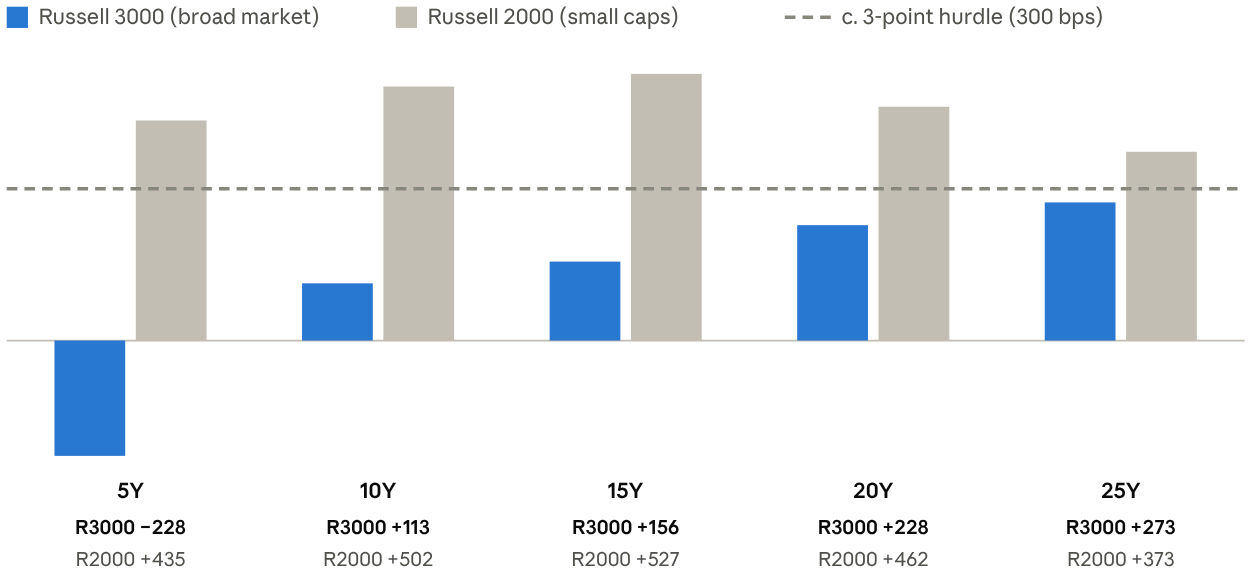
3. Findings

3.1 PE beat public markets over 25 years, but not by the margin investors require

Over 25 years, PE returned 12.76% a year, against 8.74% for the S&P 500, 9.65% for US small caps (bottom decile) and 6.18% for Baa bonds. Measured like-for-like, the premium depends on the benchmark: 2.7 points a year against the Russell 3000, below the c. 3-point hurdle many pension funds use, and 3.7 points against the Russell 2000.

PE clears the 3-point hurdle against small caps, not against the broad market

PE value-add vs mPME by horizon, bps a year, to 31 Mar 2026 · 1–3 year horizons omitted (lagged NAVs)



Cambridge Associates, US PE Benchmark Book Q1 2026: PE value-add vs Russell 3000 and Russell 2000 mPME, to 31 Mar 2026

The Russell 2000 is closer in size to buyout targets, yet the premium widens against it because US small caps lagged large caps over the period. Neither index adjusts for leverage: buyouts carry more debt than listed companies, so a risk-matched premium

would be lower. Over 1 to 3 years, PE trailed the Russell 3000 by 10.6 to 12.0 points a year, but these figures mostly reflect lagged NAV appraisals and are not used in the conclusions.

A naive comparison of PE's IRR with an index CAGR adds a further error. The 2019 report isolates the effect of the method alone, against the same index: the naive comparison misstates the premium by up to 1.5 points a year, in either direction.

Horizon (to 30 Sep 2019)	Naive gap vs S&P 500	mPME value-add vs S&P 500	Naive minus mPME
15 years	+443 bps	+400 bps	+43 bps
20 years	+538 bps	+390 bps	+148 bps
25 years	+352 bps	+431 bps	−79 bps

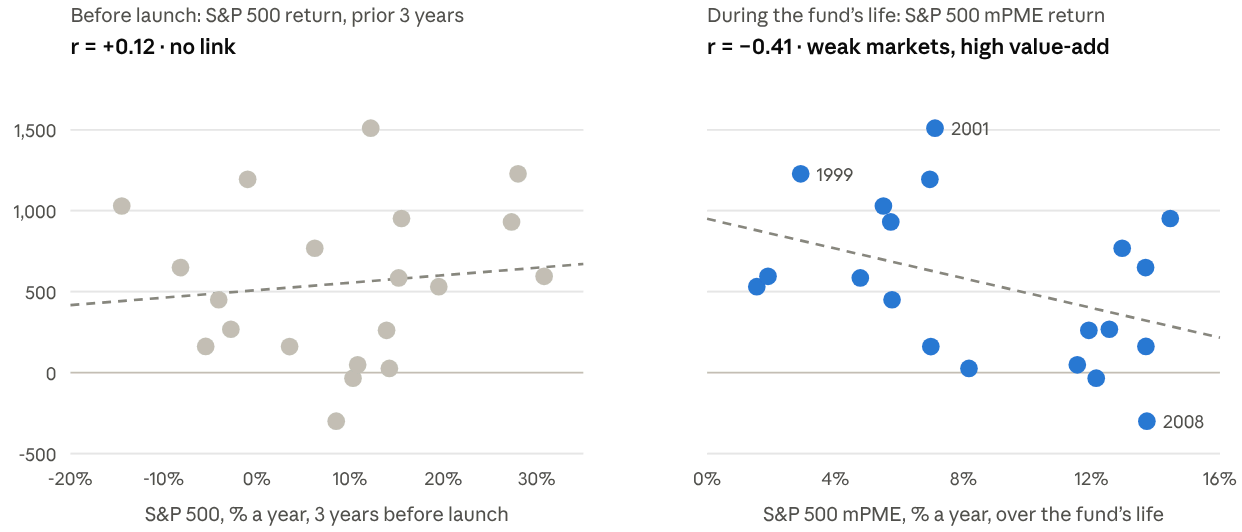
Source: Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019, pp. 7–8.

3.2 Vintage matters, but good vintages could not be picked in advance

Value-add over the S&P 500 mPME ranged from −3.0 to +15.1 points a year across the 1994–2013 vintages. Funds raised in 2001–2003, after the S&P 500's 2000–2002 decline, beat it by 12.4 points a year on average; funds raised in 2005–2008 roughly matched it (−0.4 points).

Value-add tracks public markets during a fund's life, not before its launch

Each dot is a vintage (1994–2013); vertical axis: PE value-add vs S&P 500 mPME, bps a year

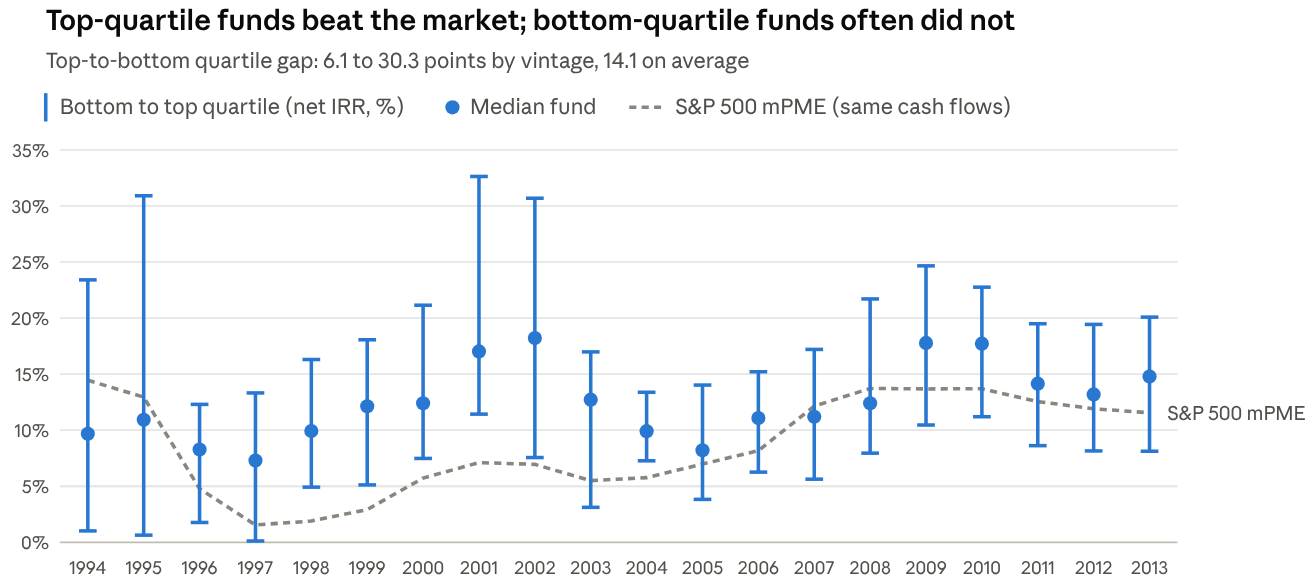


Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019 (value-add, mPME, vintages 1994–2013); A. Damodaran, NYU Stern (S&P 500 returns before launch)

What drives the gap is how public markets perform during a fund's life, not before it starts. Value-add is highest when the S&P 500 equivalent earned little (correlation −0.41), as for the 1999–2003 vintages, which include funds raised at the peak of the dot-com bubble. The S&P 500's return in the three years before launch shows no link with value-add (correlation +0.12). Recent market moves would not have helped an investor pick good vintages, which supports committing steadily every year.

3.3 Dispersion across managers is a risk, not an edge

Within a vintage, fund outcomes vary widely. On average across 1994–2013, the top-quartile threshold beat the S&P 500 mPME by 11.5 points a year and the median fund by 3.7 points, while the bottom-quartile threshold trailed it by 2.7 points.



Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019: since-inception net IRR by vintage, 1994–2013, as of 30 Sep 2019

The bottom-quartile threshold fell below the pooled S&P 500 equivalent in 14 of 20 vintages. Quartile thresholds are fund-level IRRs while the mPME is computed on pooled cash flows, so this comparison is approximate. Dispersion would be an opportunity only if good managers could be identified in advance. For buyout funds raised since 2000, past top-quartile performance has been a weak predictor of future results (Harris et al., 2020), so for a typical investor dispersion is mainly a risk.

4. What this means for investors

On average, US PE has not cleared the return hurdle investors set for it against the broad market, and its margin over small caps is thin before any leverage adjustment. PE pays off mainly for investors with privileged access to top-quartile managers, which most investors lack. Three implications follow:

1. **Access is the real question.** An average allocation to US PE has earned less than the premium needed to compensate for illiquidity. The case for PE rests on access to the best managers, not on the asset class itself.
2. **Commit steadily rather than time vintages.** Good vintages could not be identified from recent market moves, so a fixed annual commitment pace is the realistic way to capture them.

3. **Benchmark like-for-like, against the right index.** A naive IRR comparison can misstate the premium by up to 1.5 points a year, and switching from the Russell 3000 to the Russell 2000 moves it by about 1 point.

In short, PE's edge sits with the managers, not with the asset class. It pays off for investors who can access top-quartile funds and keep committing through the cycle.

5. Limitations

The conclusions rest on public, aggregated data and should be read with eight caveats:

- **Valuation smoothing:** PE NAVs are quarterly appraisals, not market prices. Recent returns (1 to 5 years) depend heavily on these estimates, and reported PE volatility is understated.
- **No risk adjustment:** buyouts are leveraged. Part of the premium may reward extra risk rather than manager skill.
- **Data access:** the latest free vintage data dates from September 2019, so vintages after 2013 are excluded as immature. Some vintages hold only 31 to 35 funds (2001–2003 and 2009–2010).
- **Definitions differ:** the 2019 data use CA's legacy definition (including mezzanine and energy funds) and define vintage by legal inception; CA's current index (2026) covers buyout and growth funds only.
- **Timing mismatch:** PE horizon returns run to March 2026, public returns to December 2025.
- **Benchmark mismatch:** the 2026 report gives the mPME vs the Russell 3000 but not vs the S&P 500, while annual public returns are only available for the S&P 500. Naive and mPME figures therefore differ by both method and index.
- **Reporting bias:** CA's database relies on funds reporting their results. Funds can be added or removed over time, so past returns may change between editions.
- **Short horizons:** 1- to 3-year PE returns rely on lagged NAV appraisals and are not used in the conclusions.

Sources

- Cambridge Associates, US Private Equity Benchmark Book, Q1 2026, as of 31 Mar 2026.

- Cambridge Associates, US Private Equity (Legacy Definition) Benchmark Book, Q3 2019, as of 30 Sep 2019.
- A. Damodaran, Historical Returns on Stocks, Bonds and Bills, 1928–2025, NYU Stern, updated 5 Jan 2026.
- Harris, Jenkinson, Kaplan & Stucke, *Has Persistence Persisted in Private Equity? Evidence from Buyout and Venture Capital Funds*, NBER Working Paper 28109, 2020.
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- Institutional Investor, *Public pensions frequently change their private equity benchmarks. Why it matters*.
- Kaplan & Schoar, *Private Equity Performance: Returns, Persistence, and Capital Flows*, Journal of Finance 60(4), 2005.
- Harris, Jenkinson & Kaplan, *Private Equity Performance: What Do We Know?*, Journal of Finance 69(5), 2014.

Glossary

Players

- **Private equity (PE):** investment in companies that are not listed on a stock exchange, usually by buying a majority stake, improving the business and selling it a few years later.
- **Buyout (LBO):** the purchase of a company financed partly with debt, to increase the return on the money invested.
- **Fund:** a pool of money raised by a PE firm for a limited life (about 10 years), used to buy and later sell several companies.
- **GP (General Partner):** the PE firm that manages the fund and chooses the investments (e.g. KKR, Ardian). Also called the *manager*.
- **LP (Limited Partner):** an investor that puts money into a PE fund (e.g. a pension fund, an insurer).
- **Vintage year:** the year a fund starts investing. Funds of the same vintage invest in the same market conditions.
- **Capital call / distribution:** a capital call is money a fund requests from its LPs to make an investment; a distribution is cash a fund returns to its LPs, usually after selling a company.
- **Carried interest:** the share of a fund's profits (usually about 20%) kept by the GP when the fund performs well.

Returns

- **IRR (Internal Rate of Return):** the annual return of a PE fund, taking into account *when* money is invested and returned. Also called a *money-weighted* return.
- **CAGR (Compound Annual Growth Rate):** the average annual return of an investment held throughout a period, with gains reinvested. Also called a *time-weighted* return.
- **Pooled return:** the combined return of all funds in a group, as if they were one large fund. Bigger funds weigh more.
- **Basis point (bp):** one hundredth of a percentage point. 100 bps = 1 point; 273 bps = 2.73 points.
- **Quartile:** one of four equal groups when funds are ranked from best to worst. *Top quartile* = the best 25%; *bottom quartile* = the worst 25%.
- **NAV (Net Asset Value):** the estimated value of the companies a fund still owns. It is an appraisal, not a market price.

Comparing PE with public markets

- **Public markets:** stocks and bonds traded on an exchange, which can be bought or sold at any time.
- **S&P 500:** an index of the 500 largest US listed companies.
- **Russell 3000:** an index of the 3,000 largest US listed companies, covering almost the whole US stock market.
- **Russell 2000:** an index of the 2,000 smallest companies in the Russell 3000.
- **US small caps (bottom decile):** the smallest 10% of US listed companies by market value, as used in Damodaran's data.
- **Baa corporate bonds:** bonds issued by US companies with a medium credit rating (Moody's Baa), just above high-yield status.
- **T-bills / 10Y T-bonds:** US government debt maturing in 3 months / 10 years.
- **Naive comparison:** subtracting a stock index's CAGR from a PE fund's IRR. Simple but misleading, because the two returns are not measured the same way.
- **PME (Public Market Equivalent):** a method that invests a PE fund's cash flows in a stock index, on the same dates, to compare like with like.
- **mPME (modified PME):** Cambridge Associates' version of the PME, which sells the same *proportion* of the index at each fund distribution, so the shadow portfolio never turns negative.

- **Value-add:** the PE fund's IRR minus the return of its mPME. A positive value-add means PE beat the stock market.

Context

- **Illiquidity premium:** the extra return investors expect for locking up their money for years. Many pension funds expect PE to beat stock markets by about 3 points a year.
- **Dot-com crash (2000–2002):** the collapse of internet and technology stocks after a speculative bubble.
- **GFC (Global Financial Crisis, 2007–2009):** the financial crisis triggered by US subprime mortgages, marked by the failure of Lehman Brothers in 2008.